Electronic Spreadsheet (Advanced)

The big idea

LibreOffice Calc can do far more than add numbers. In Class 10 you learn to **combine data** from many sheets (consolidate), **summarise** data with subtotals, ask **"what if?"** questions with scenarios, goal seek and solver, **link** sheets and files, **share and track changes**, and automate repeated work with **macros**.

Consolidating data

Consolidation combines data from several ranges (often on different sheets) into one summary.

1. Open the sheet where the result should go.
2. **Data > Consolidate**.
3. Choose a **Function** (Sum, Count, Average, Max, Min, Product, Count Numbers, StDev, Var…).
4. In **Source data ranges**, select each range and click **Add**.
5. In **Copy results to**, select the target cell.
6. Under **Options**, tick **Row labels / Column labels** if needed and **Link to source data** so the result updates when the source changes. Click **OK**.

Subtotals

Data > Subtotals inserts sub-totals for groups of rows.

1. **Sort** the data first by the column you will group on (Data > Sort).
2. Select the data → **Data > Subtotals**.

3. In the **1st Group** tab: *Group by* (e.g. Region), *Calculate subtotals for* (e.g. Sales), *Use function* (Sum, Average, Count…).
4. Click **OK**. Calc shows outline buttons (1, 2, 3) on the left to show/hide detail.

SORT BEFORE SUBTOTALS

If the grouping column is not sorted, the same group appears in many places and you get many small, wrong subtotals.

What-if analysis

Tool	Menu	Use
Scenarios	**Tools > Scenarios**	Save different sets of input values (e.g. best/likely/worst case) and switch between them using the Navigator
Goal Seek	**Tools > Goal Seek**	Find the **one input** needed to reach a desired result
Solver	**Tools > Solver**	Find the best values of **several inputs** with constraints (maximise/minimise)
Multiple Operations	**Data > Multiple Operations**	Build a table showing results for many input values

GOAL SEEK

A student has 70, 80 and 85 in three tests (B1:B3) and the fourth test in B4. Average in B5: `=AVERAGE(B1:B4)`. What score in B4 gives an average of 80? **Tools > Goal Seek** → Formula cell: B5, Target value: 80, Variable cell: B4 → OK. Calc answers **85**.

Linking data

- Reference another sheet: `=Sheet2.B4` (Calc uses a **dot**; with quotes for names with spaces: `='Term 1'.B4`). Or type `=`, click the other sheet's cell, press Enter.
- Reference another file: `='file:///home/user/marks.ods'#$Sheet1.A1`.
- **Hyperlinks**: **Insert > Hyperlink** (Ctrl+K) to link to a website, file, or a cell/sheet in the same document.
- **Sheet > Link to External Data** brings in a table from a web page or another file.

Sharing and reviewing

- **Share**: **Tools > Share Spreadsheet** – tick *Share this spreadsheet with other users*, so several people can edit it.
- **Record changes**: **Edit > Track Changes > Record** (Ctrl+Shift+C); changes are highlighted with author and time.
- **Accept/reject**: **Edit > Track Changes > Manage**.
- **Comments**: **Insert > Comment** (Ctrl+Alt+C).
- **Compare**: **Edit > Track Changes > Compare Document**; **Merge Document** combines copies.

Macros

A **macro** is a saved sequence of commands or keystrokes that can be replayed to automate repetitive tasks.

1. Enable recording: **Tools > Options > LibreOffice > Advanced > Enable macro recording**.
2. **Tools > Macros > Record Macro**, do the actions, click **Stop Recording**.
3. Choose a library/module (e.g. in **Standard**), give a name, **Save**.
4. Run: **Tools > Macros > Run Macro**, select it, **Run**.

Macros are written in **LibreOffice Basic** and can also be used as custom functions in formulas.

ONE-INPUT VS MANY-INPUT

Goal Seek = one variable, one target. Solver = many variables with constraints. Examiners love this difference.

Quick recap

- Data > Consolidate merges ranges with a function; Link to source data keeps it live.
- Data > Subtotals after sorting; outline levels 1–2–3.
- Scenarios, Goal Seek, Solver, Multiple Operations = what-if tools.
- `=Sheet2.A1` links sheets; Ctrl+K inserts hyperlink.
- Tools > Share Spreadsheet; Edit > Track Changes > Record.
- Macros: Tools > Macros > Record Macro / Run Macro.

Practice questions

1. Which menu option combines data from several ranges into one summary?
   (a) Data > Subtotals (b) Data > Consolidate (c) Tools > Goal Seek (d) Sheet > Insert Sheet
2. Before applying subtotals, the data should be
   (a) protected (b) shared (c) sorted by the grouping column (d) converted to a chart
3. Which tool finds the value of one input cell needed to reach a target result?
   (a) Solver (b) Scenarios (c) Consolidate (d) Goal Seek
4. In LibreOffice Calc, the correct reference to cell B4 on Sheet2 is
   (a) =Sheet2.B4 (b) =Sheet2!B4 (c) =B4:Sheet2 (d) =Sheet2#B4
5. Which feature lets you save best-case and worst-case sets of input values and switch between them?
   (a) Macros (b) Scenarios (c) Hyperlinks (d) Subtotals
6. The shortcut to start recording changes in a Calc document is
   (a) Ctrl+Shift+F4 (b) Ctrl+K (c) Ctrl+Shift+C (d) Ctrl+F12
7. A macro is best described as
   (a) a large chart (b) a type of cell reference (c) a protected sheet (d) a recorded set of actions that can be replayed
8. Differentiate between Goal Seek and Solver.
9. Write the steps to record and run a macro in LibreOffice Calc.

Answers: 1. b 2. c 3. d 4. a 5. b 6. c 7. d 8. see notes 9. see notes